

# Schedule premaster PMTPS - Probability & Statistics (2026-2027)

| Learning task                                                              | Book sections (6th SI version)                                  | Lecture notes                              |
|----------------------------------------------------------------------------|-----------------------------------------------------------------|--------------------------------------------|
| <b>The Role of Statistics in Engineering, Introduction to Probability</b>  |                                                                 |                                            |
| 1                                                                          | Sections 1-1, 1-2, 1-3, 1-4<br>Sections 2-1, 2-2, 2-3, 2-4, 2-5 | PMTPS_STAT_Ch1.pdf<br>PMTPS_STAT_Ch2#1.pdf |
| 2                                                                          | Sections 2-6, 2-7, 2-8                                          | PMTPS_STAT_Ch2#2.pdf                       |
| <b>Discrete and Continuous Random Variables; Probability Distributions</b> |                                                                 |                                            |
| 3                                                                          | Sections 3-1, 3-2, 3-3, 3-4, 3-5                                | PMTPS_STAT_Ch3#1.pdf                       |
| 4                                                                          | Sections 3-6, 3-7.1, 3-9                                        | PMTPS_STAT_Ch3#2.pdf                       |
| 5                                                                          | Sections 4-1, 4-2, 4-3, 4-4, 4-5                                | PMTPS_STAT_Ch4#1.pdf                       |
| 6                                                                          | Sections 4-6, 4-8                                               | PMTPS_STAT_Ch4#2.pdf                       |
| <b>Joint Probability Distributions, Central Limit Theorem</b>              |                                                                 |                                            |
| 7                                                                          | Sections 5-1, 5-2 (discrete)                                    | PMTPS_STAT_Ch5#1.pdf                       |
| 8                                                                          | Sections 5-1, 5-2 (continuous), 5-3.2                           | PMTPS_STAT_Ch5#2.pdf                       |
| 9                                                                          | Sections 5-4                                                    | PMTPS_STAT_Ch5#3.pdf                       |
| <b>10</b>                                                                  | <b>Sections 7-1, 7-2</b>                                        | <b>PMTPS_STAT_Ch7#1.pdf</b>                |
| <b>Point and Interval Estimation</b>                                       |                                                                 |                                            |
| 11                                                                         | Section 7-3                                                     | PMTPS_STAT_Ch7#2.pdf                       |
| 12                                                                         | Section 7-4.2 (until: Invariance Property)                      | PMTPS_STAT_Ch7#3.pdf                       |
| 13                                                                         | Section 8-1                                                     | PMTPS_STAT_Ch8#1.pdf                       |
| 14                                                                         | Sections 6-7, 8-2                                               | PMTPS_STAT_Ch8#2.pdf                       |
| <b>Tests of Hypotheses for a Single Sample</b>                             |                                                                 |                                            |
| 15                                                                         | Sections 9-1.1, 9-1.2, 9-1.3, 9-1.4, 9-2.1                      | PMTPS_STAT_Ch9#1.pdf                       |
| 16                                                                         | Sections 9-1.5, 9-1.6, 9-3.1                                    | PMTPS_STAT_Ch9#2.pdf                       |
| 17                                                                         | Sections 9-2.2                                                  | PMTPS_STAT_Ch9#3.pdf                       |
| 18                                                                         | Sections 4-9, 9-7                                               | PMTPS_STAT_Ch9#4.pdf                       |
| <b>Simple Linear Regression and Correlation</b>                            |                                                                 |                                            |
| 19                                                                         | Sections 11-1, 11-2                                             | PMTPS_STAT_Ch11#1.pdf                      |
| 20                                                                         | Section 11-8                                                    | PMTPS_STAT_Ch11#2.pdf                      |